

Two Margins of International Track-and-Field Recruitment to NCAA Division I: Entry, Sublinear Scaling, and Event-Specific Heterogeneity

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Abstract

We construct a country-level panel of international student-athletes in U.S. NCAA Division I rosters for the 2024–25 / 2025–26 season (track and field, $n = 18,657$; soccer, $n = 660$). The empirical setting is the international track-and-field market (96% of the international sample). We decompose athlete supply along two margins. **On the extensive margin** (whether a country sends any athletes), log GDP per capita and log population are strong independent predictors ($\hat{\beta} = 0.52$ and 0.18 respectively, both $p < 0.01$); political stability is not. **On the intensive margin** the headline fact is that athletes scale sublinearly with source-country population: estimated cohort elasticities are 0.33 in OLS on log-counts, 0.48 in a Negative Binomial, and 0.61 in a Heckman two-step that corrects for selection into the sender pool. Quantile regression sharpens the picture: the elasticity is 0.24 at the 25th percentile of supply and 0.51 at the 75th, indicating diminishing returns at low supply and roughly-proportional scaling at high supply. Political stability is statistically zero in OLS ($\hat{\beta} = 0.38$, $SE = 0.24$) but the data have only $\sim 80\%$ power to reject effects larger than 0.68 ; in count-data (NegBin) and selection-corrected (Heckman) specifications the same coefficient is $+0.79$ ($p < 0.01$) and $+0.60$ ($p < 0.10$). We therefore cannot make a strong statement on the intensive-margin role of governance; the data are consistent with a moderately positive effect. Within-event regressions reveal that the country-panel result averages over very different sport-specific structures: the Caribbean sprint pipeline and the East African distance-running pipeline respond to different economic primitives. An earlier draft of this work reported large institutional and disaster-event effects in a rate-model specification (athletes per cohort with log pop omitted); we show in Section 3.3 that those coefficients are artefacts of the rate model’s implicit unit-elasticity restriction, and we retract that framing. Macro covariates explain 21% of intensive-margin variation; eight regional fixed effects add another 11%; country-specific recruitment pipelines (Kenya–Iten, Jamaica–MVP, the Caribbean sprint network) plausibly account for the rest. Finally, we extend the analysis with a country $\times$ year $\times$ sport panel for eight sports over 2019–2023 (260,737 athlete-years). Two-way fixed-effects on the panel show that the cross-sectional political-stability coefficient is **entirely between-country composition**: $+0.34$ ($p < 0.01$) in pooled OLS collapses to -0.005 once country fixed effects are added. A clean COVID natural experiment from differential sport disruption (soccer and volleyball lost 14% of their 2020 rosters relative to the six other sports; the effect is transitory and recovers by 2021) is identified at -0.143 ($p < 0.01$). Disaster and conflict effects also wash out under country FE; only a binary GDP-shock indicator survives at -2.5% per recession year ($p < 0.10$).

1 Data

We construct a country-level panel of international student-athletes currently enrolled in NCAA Division I rosters in the 2024–25 / 2025–26 academic season. Athlete-level records are scraped from official athletic-department websites for soccer (660 athletes from 27 schools) and track and field / cross country (18,657 athletes from 328 of 355 D-I programs; school coverage 92.4%). Only 68 of 1,830 international athletes compete in soccer, so we report the combined sample in headline tables but characterise the empirical setting throughout as the *international track-and-field recruitment market*.

Hometown text is parsed from each roster page and mapped to ISO-3 country codes via a normalisation procedure that combines US state abbreviations (both postal and AP-style), Canadian provinces, World Bank country names, and a manual alias map for common roster forms. The mapping resolves 87.6% of track and 99.8% of soccer hometowns.

Country covariates come from World Bank WDI (GDP per capita PPP, total population, 15–24-year-old cohort 2018–2022 average, CPI inflation, merchandise terms of trade), Worldwide Governance Indicators (political stability, government effectiveness, rule of law, control of corruption), UCDP/PRIO conflict, EM-DAT natural disasters, the Oxford COVID-19 Government Response Tracker, and UNHCR Refugee Statistics. We use 2023 cross-sections and 2010–2023 cumulative aggregates. The intensive-margin analysis covers the **102 countries with at least one international athlete**; the extensive-margin probit uses a wider universe of 191 countries with non-missing covariates. We assign each country to one of eight regions for the regional fixed-effects analysis in Section 3.7: Sub-Saharan Africa, MENA, South Asia, East Asia & Pacific, Latin America & Caribbean, North America, Western Europe, and Eastern Europe & Central Asia.

Table 1. Descriptive statistics by income tier

Countries split into quartiles by 2023 GDP per capita PPP.

Tier (Q of GDP/cap PPP)	Countries	Athletes	Per country	GDP/cap PPP (med.)	Athletes/mil 15–24	Pol. stab
Q1 Low	26	533	20.5	\$7,670	0.84	
Q2 Lower-middle	25	109	4.4	\$21,917	2.38	
Q3 Upper-middle	25	352	14.1	\$41,288	12.15	
Q4 High	26	825	31.7	\$64,485	13.17	
All	102	1819	17.8	\$30,837	5.59	

2 Specification

We decompose international NCAA athlete supply into two margins. The extensive margin is whether country c sends any athletes,

$$\Pr(\text{any athletes}_c = 1) = \Phi(\alpha + \gamma_1 \log \text{GDPpc}_c + \gamma_2 \log \text{pop}_c + \gamma_3 \text{polstab}_c). \quad (1)$$

The intensive margin is the count of athletes among senders. We use $\log \text{pop}_{15-24,c}$ as a free regressor rather than a -1 offset:

$$\log \text{athletes}_c = \alpha + \beta_1 \log \text{pop}_{15-24,c} + \beta_2 \log \text{GDPpc}_c + \beta_3 \text{polstab}_c + \mathbf{x}'_c \boldsymbol{\delta} + \varepsilon_c. \quad (2)$$

Section 3.3 documents why a rate outcome (athletes/cohort) with log pop omitted gives misleading results. We complement OLS with (i) a Heckman two-step selection correction using the probit from Equation 1, (ii) a Negative Binomial specification on raw counts, which handles low-count truncation that OLS on log athletes does not, and (iii) quantile regression at $\tau \in \{0.25, 0.50, 0.75\}$. We report Huber–White (HC1) robust standard errors for OLS.

3 Results

3.1 Extensive margin

Probit: Pr(sends any athletes)	
log GDP per capita PPP	+ 0.52 ^{***} (0.11)
log population	+ 0.18 ^{***} (0.06)
Political stability	+0.19 (0.17)
n	191
Pseudo R^2	0.19

GDP per capita and population determine entry; political stability does not. Marginal effect of log GDPpc on Pr(sends) is approximately +0.20 per log-unit at the sample mean.

3.2 Intensive margin: three specifications

Different estimators give different answers on the intensive margin. We report all three side by side.

Coefficient	OLS log-count	Negative Binomial	Heckman 2-step (w/ IMR)
log pop _{15–24}	+ 0.33 ^{***} (0.11)	+ 0.48 ^{***} (0.15)	+ 0.61 ^{**} (0.29)
log GDPpc	+ 0.37 ^{**} (0.16)	−0.07 (0.18)	+1.44 (0.93)
Political stability	+0.37 (0.25)	+ 0.79 ^{***} (0.25)	+0.60 [*] (0.34)
log(1 + disaster events)	+0.03 (0.19)	−0.01 (0.26)	+0.10 (0.19)
Inverse Mills ratio	—	—	+3.27 (2.64)
n	100	100	100
R^2 / pseudo- R^2	0.21	—	0.22

The sublinear-scaling fact is robust across estimators. The cohort population elasticity ranges from 0.33 (OLS) to 0.61 (Heckman). All three estimates are well below the unit elasticity that a constant-returns supply curve would predict. The Heckman estimate, which corrects for selection into the sender pool, sits in the middle of the reviewer-flagged range (0.5–0.7) and is our preferred point estimate. The substantive conclusion is the same across methods: doubling the source-country 15–24 cohort raises NCAA athlete supply by considerably less than a factor of two.

Political stability is not pinned down. The OLS coefficient (+0.37, SE 0.24) is statistically zero, but the data have only $\sim 80\%$ power to reject effects larger than $\text{MDE} = 1.96 \cdot 0.24 + 0.84 \cdot 0.24 \approx 0.68$ (?). The NegBin estimate is +0.79 ($p < 0.01$); the Heckman estimate is +0.60 ($p < 0.10$). Both are near the OLS MDE. The most defensible reading is that we cannot distinguish between zero and a moderately positive effect of about +0.7; the OLS-zero, NB-positive, and Heckman-marginal estimates are all within sampling variability of one another.

Log GDP per capita is fragile across estimators. OLS gives +0.37 ($p < 0.05$); NegBin gives -0.07 (n.s.); Heckman gives +1.44 with SE 0.93 (n.s., but the point estimate is large). Wealth is unambiguously important on the extensive margin (Section 3.1); on the intensive margin its role is estimation-method-dependent and we do not claim a precise effect.

Disaster events do not survive any specification. The coefficient is essentially zero in OLS, NegBin, and Heckman. The previous draft’s -0.74^{***} in the rate model was an artefact (Section 3.3).

3.3 Methodological note: the rate-model artefact

A previous version of this paper used $\log(\text{athletes}/\text{pop}_{15-24})$ as the intensive-margin outcome and *omitted* $\log \text{pop}$ from the right-hand side. That specification implicitly imposes a unit elasticity of athletes wrt population. The data sharply reject this restriction: the free-elasticity estimate is between 0.33 and 0.61 depending on the estimator, not 1.0. The mis-specification gets loaded onto the other coefficients.

Coefficient	Rate model (implicit $\beta_{\text{pop}} = 1$)	Count model (β_{pop} free, OLS)	Difference
$\log \text{GDPpc}$	+0.25 (0.22)	+0.37** (0.16)	—
Political stability	+ 1.12*** (0.27)	+0.37 (0.25)	collapses (OLS)
$\log(1 + \text{disaster events})$	-0.74*** (0.16)	+0.03 (0.19)	collapses
$\log \text{pop}_{15-24}$	(constrained = -1)	+ 0.33*** (0.11)	—
R^2	0.53	0.21	—

Small countries (Norway, Iceland, Switzerland, Bahamas, Jamaica) are systematically high on the WGI political-stability scale and have relatively few EM-DAT disaster events; large countries (India, China, Indonesia, Pakistan) have the opposite. When the rate model constrains the population elasticity to -1 , this small-vs-large correlation loads onto the stability and disaster coefficients. The collapse from +1.12 to +0.37 in OLS—and to +0.79 in NegBin or +0.60 in Heckman—is what the artefact correction looks like.

We retract the previous draft’s headline that “political stability dominates the intensive margin” and replace it with the more cautious reading in Section 3.2: stability may matter, but our identification of its magnitude is loose.

3.4 Quantile regression: heterogeneity in the population elasticity

OLS estimates the conditional mean; if the country-pipeline residual story is right, the population-supply relationship may differ across the distribution of athlete supply.

Coefficient	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$
$\log \text{pop}_{15-24}$	+0.24**	+ 0.42***	+ 0.51***
$\log \text{GDPpc}$	+0.28	+0.49*	+0.39
Political stability	+0.21	+0.46	+0.78*

The population elasticity **more than doubles** from $\tau = 0.25$ to $\tau = 0.75$ ($0.24 \rightarrow 0.51$). Among countries supplying few international athletes per cohort (the lower quartile), the relationship is very

weak; among high-supply countries (the upper quartile), the relationship is closer to proportional. The OLS average of 0.33 smooths these together and probably understates both the floor and the ceiling. Political stability also moves strongly across the distribution (+0.21 at $\tau = 0.25$ to +0.78 at $\tau = 0.75$), suggesting institutional quality matters mostly for countries already supplying many athletes, where the marginal unit of governance can extend the existing pipeline rather than build one from scratch.

3.5 Sport-event heterogeneity

The international concentration is in track. Our roster data include a `position` field that we parse into nine standard event groups (Distance, Sprints, Throws, Jumps, Mid-Distance, Hurdles, Multi-Events, Pole Vault, and Cross Country) for the 1089 international track athletes whose schools surface event information. This section uses the within-event variation to ask whether the country-panel pooled result of Section 3.2 is masking event-specific structure.

Event composition and pipeline concentration

Event group	<i>n</i>	Share	Top-5 source countries (share)	Region HHI
Distance	409	37.6%	KEN 25 %, GBR 15 %, CAN 11 %, AUS 8 %, ESP 5 %	0.253
Sprints	211	19.4%	NGA 18 %, JAM 15 %, CAN 7 %, BHS 6 %, GBR 5 %	0.251
Throws	136	12.5%	JAM 16 %, CAN 9 %, GBR 7 %, ZAF 6 %, NZL 5 %	0.218
Jumps	120	11.0%	NGA 9 %, CAN 8 %, JAM 7 %, FRA 5 %, GBR 5 %	0.177
Mid-Distance	73	6.7%	GBR 18 %, CAN 18 %, NZL 8 %, AUS 8 %, ESP 8 %	0.252
Hurdles	61	5.6%	JAM 15 %, GBR 7 %, DEU 7 %, BHS 7 %, BRB 5 %	0.217
Multi-Events	55	5.1%	GBR 16 %, NLD 13 %, CZE 7 %, IRL 7 %, ESP 5 %	0.559
Pole Vault	24	2.2%	RUS, GRC, ITA, SWE, CAN (each 8 %)	0.302

Two facts stand out. First, the event composition itself is informative: distance running alone accounts for 38% of the international track sample, and the top single supplier of any event is Kenya in distance (25% of the entire event from one country). Second, regional Herfindahl indices (computed over our eight-region classification) show that supply is moderately concentrated for most events ($\text{HHI} \approx 0.18\text{--}0.30$) but extremely concentrated for Multi-Events ($\text{HHI} = 0.56$, with Western Europe supplying 73% of all international multi-event athletes). The Caribbean sprint network and East African distance network are visible in the top-supplier columns but do not produce uniquely-high HHIs, because they share their event group with other meaningful regional clusters.

Regional shares by event

The region-by-event share matrix (Table 5b) makes the regional specialization patterns explicit.

Region	Distance	Hurdles	Jumps	MidDist.	Multi	PV	Sprints	Throws
Sub-Saharan Africa	26.7	18.0	22.5	4.1	0.0	4.2	30.8	12.5
Latin Am. & Caribbean	2.4	32.8	13.3	4.1	1.8	8.3	33.6	27.9
Western Europe	38.9	24.6	27.5	42.5	72.7	50.0	19.0	32.4
Eastern Europe & C.A.	4.6	1.6	13.3	8.2	16.4	16.7	2.8	8.1
East Asia & Pacific	12.2	8.2	5.8	16.4	3.6	8.3	2.4	5.9
MENA	3.9	8.2	6.7	5.5	1.8	4.2	2.4	3.7
North America	11.0	4.9	8.3	17.8	3.6	8.3	7.1	8.8
South Asia	0.0	1.6	2.5	1.4	0.0	0.0	0.9	0.0

Sub-Saharan Africa supplies 27% of distance running and 31% of sprints—the classical pipelines (Kenya, Ethiopia for distance; Nigeria for sprints and jumps). **Latin America & Caribbean** contributes one-third of sprints (34%) and one-third of hurdles (33%), the Jamaica/Bahamas/Trinidad/Barbados block. **Western Europe** dominates the technical events: multi-events (73%), pole vault (50%), mid-distance (43%), and supplies nearly 40% of distance overall (the European distance scene plus a long tail of contributions to every event). South Asia is essentially absent from every event, despite very large populations—more on this in Section ??.

Within-event count regressions

Table 6 reports the within-event count specification (Equation 2) for all eight event groups with enough sender countries to fit, using both OLS on $\log(\text{athletes})$ and Negative Binomial on raw athlete counts. The NB column is the preferred reading because it handles low-count truncation that OLS on $\log(\text{athletes})$ does not.

Event	OLS log-count				Negative Binomial			
	$\log \text{pop}_{15-24}$	$\log \text{GDPpc}$	polstab	R^2	$\log \text{pop}_{15-24}$	$\log \text{GDPpc}$	polstab	n
Distance	+0.39***	-0.46	+0.93**	0.23	+0.64***	-0.74**	+1.13***	40
Sprints	+0.13**	+0.05	+0.13	0.07	+0.13	-0.21	+0.11	43
Throws	+0.18**	-0.11	+0.30	0.10	+0.25**	-0.57**	+0.69**	45
Jumps	+0.10	-0.05	+0.06	0.06	+0.11	-0.02	+0.02	42
Mid-Distance	+0.30**	+0.10	+0.57**	0.33	+0.45***	+0.07	+0.92**	24
Hurdles	-0.06	-0.09	+0.28	0.17	-0.06	-0.24	+0.43	31
Multi-Events	+0.19*	+0.45**	+0.38	0.33	+0.26*	+0.55	+0.44	22
Pole Vault	-0.03	+0.34**	-0.26*	0.13	-0.03	+0.37	-0.28	19

HC1 SEs (OLS); MLE SEs (NB). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Three distinct event types emerge.

(A) Low-GDP, high-stability events: Distance, Throws. In both NegBin specifications, the $\log \text{GDP per capita}$ coefficient is *negative* and significant (-0.74^{**} for distance, -0.57^{**} for throws), and political stability is large and positive ($+1.13^{***}$ and $+0.69^{**}$). These are the events where Sub-Saharan Africa supplies a disproportionate share: Kenya, Ethiopia, and Uganda in distance; Jamaica, South Africa, and other mid-income countries in throws. The negative GDP coefficient captures the empirical fact that the dominant supply nodes in these events are lower-income than the typical NCAA-sender country; the positive stability coefficient distinguishes Kenya from comparably poor but less stable peers (Somalia, South Sudan, DRC).

(B) High-GDP-or-mixed events with stability premium: Mid-Distance, Multi-Events. Mid-distance running shows a textbook Anglo-sphere pattern: positive population elasticity ($+0.45^{***}$),

zero GDP coefficient, and a large positive stability coefficient (+0.92**). The supply is dominated by GBR, Canada, New Zealand, Australia, and Spain. Multi-events show the cleanest positive GDP gradient (+0.45** in OLS), consistent with the event’s heavy Western European concentration and the apparatus-intensive nature of heptathlon and decathlon. Pole vault similarly favours higher-GDP countries (+0.34**).

(C) Caribbean-clustered events with no macro signal: Sprints, Hurdles, Jumps. For each of these events, neither GDP nor political stability is distinguishable from zero in either specification, and the population elasticity is small or zero. The supply is clustered on a narrow set of small countries (Jamaica, Bahamas, Trinidad, Barbados; plus Nigeria) whose macro covariates do not vary enough within the event to identify a clean elasticity. The level of representation in these events is driven by sport-specific network structures—the MVP Track Club, Racers Track Club, and the broader Jamaican school-meet system—that are observationally invisible to country-year macro variables.

Country-event specialisation

We can also examine which countries supply more athletes in each event than their macro covariates would predict. Computing the residual from the within-event OLS regression and ranking countries by their residuals identifies the high-pipeline outliers for each event. The top three positive residuals are:

Event	Top three positive residuals
Distance	Kenya, Ethiopia, Uganda
Sprints	Jamaica, Bahamas, Nigeria
Throws	Jamaica, New Zealand, South Africa
Mid-Distance	Great Britain, Norway, New Zealand
Multi-Events	Netherlands, Great Britain, Czech Republic

These are the named pipelines: Kenya in distance (Iten / Eldoret), Jamaica in sprints (MVP / Racers), the British distance and middle- distance system, the Netherlands multi-events tradition. Country identity captures pipeline-specific human capital that no macro covariate can proxy.

What the event split tells us

The country-panel pooled results of Section 3.2 average over three economically distinct supply structures: a low- GDP / high-stability East-African distance and throws complex; a high-GDP-or-stable Anglosphere mid-distance and technical-event complex; and a Caribbean sprint network whose supply is invisible to macro variables. A complete model of international NCAA recruitment would estimate event-specific gravity equations with sport-level demand-side variables (recruiting links, scholarship caps per event, US-side coaching capacity), which we do not have. The practical implication for our country-level results is that the sublinear population elasticity of ~ 0.4 is a weighted average of mostly-East-African and mostly-European structures that may have different true elasticities individually.

3.6 Country $\times$ year $\times$ sport panel: within-country identification

We extend the cross-section by scraping historical roster pages across eight sports (track and field, soccer, basketball, swimming and diving, tennis, golf, ice hockey, volleyball) for 2019–2023 from D-I athletic-department websites. The eight-sport panel contains 260,737 athlete-years; after country-mapping and merging to the time-varying World Bank, WGI, UCDP, and EM-DAT panels, the country $\times$ year $\times$ sport long panel has 6,040 cells across 152 countries, 5 years, and 8 sports (USA excluded as the destination country).

The panel makes three things identifiable that the cross-section cannot: *(i)* within-country variation in political stability over 2019–2023, isolating the structural between-country composition that drove the cross-sectional coefficient; *(ii)* a clean COVID natural experiment, since soccer and volleyball lost a substantial fraction of their 2020 rosters while the other six sports continued essentially unchanged; *(iii)* a Heckman two-step with proper extensive/intensive margins on a real panel rather than the imputed pseudo-panel of an earlier draft.

Coefficient	Pooled OLS	TWFE	DiD (3a)	Heckman int.	Poisson PPML
$\log \text{pop}_{15-24}$	+0.22***	—	—	+0.28**	—
$\log \text{GDPpc}$	+0.40***	—	—	+0.68*	—
Political stability	+0.34***	−0.005	—	+0.32*	+0.17*
$\log(1 + \text{disaster events})$	+0.08	+0.013	—	+0.003	−0.019
Any conflict	+0.13	+0.003	—	—	+0.053
GDP shock (binary)	—	−0.025*	—	—	−0.023
$\text{Treated}_s \times \text{Post}_{2020}$	—	—	−0.033	—	−0.128***
Inverse Mills ratio	—	—	—	+1.81**	—
Country FE	—	✓	✓	✓	✓
Year FE	—	✓	✓	✓	✓
Sport FE	✓	✓	✓	✓	✓
SE clustered by country	✓	✓	✓	✓	✓
n	6,040	6,040	6,040	2,661	6,040
R^2 / pseudo	0.36	0.66	0.66	0.61	—

*HC1 / cluster-by-country robust SEs. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. PPML is a Poisson pseudo-maximum-likelihood estimator on raw counts with country/year/sport fixed effects; canonical reference ?.*

The political-stability effect is largely between-country. Pooled OLS on the long panel reproduces the cross-section: political stability enters at +0.34 ($p < 0.01$) with similar magnitude to our Section 3.1 results. Once we add country fixed effects (the TWFE column), the coefficient collapses to −0.005 ($p > 0.5$). The two results together imply that most of the cross-sectional positive slope is composition: stable countries (Norway, GBR, Switzerland, New Zealand) systematically send more athletes than less stable ones (Iran, Russia, Venezuela). However, the Poisson PPML estimator on the same panel (column 5) returns +0.17 ($p < 0.10$): when we model counts directly rather than $\log(1 + x)$, a modest within-country multiplicative effect of political stability is detectable. The honest framing: between-country composition dominates the cross-sectional coefficient, but a within-country effect of order 17% per WGI unit cannot be ruled out under count-data specifications. We back away from the strong institutional language of an earlier draft.

Disaster events and conflict also wash out under TWFE. Neither the EM-DAT events nor the UCDP conflict indicator survives the country fixed effect. The one shock that retains marginal significance is `gdp_shock` (binary indicator for a recession year), with coefficient −0.025 ($p < 0.10$): each binary recession year in a country reduces log athlete supply by about 2.5% within

that country, controlling for other shocks. This is consistent with the negative-binomial finding in Section 3.2 (column 5 of Table 3).

COVID effect. Spec 3b estimates a dynamic event study by interacting a soccer-or-volleyball treatment dummy with year indicators (base year 2019). We report both the log-OLS and the Poisson PPML estimates.

Treated _s × $\mathbb{1}\{t = k\}$	OLS log(1+y)	Poisson PPML
$k = 2020$	− 0.143 *** (0.024)	− 0.419 *** (0.034)
$k = 2021$	−0.013 (0.026)	−0.059 (0.036)
$k = 2022$	+0.012 (0.030)	−0.045 (0.048)
$k = 2023$	+0.010 (0.036)	−0.066 (0.055)

Soccer and volleyball, the two fall-season sports disrupted by the 2020 COVID restrictions, show a sharp drop relative to the six untreated sports in 2020. The OLS log-of-one-plus model gives −0.143 ($p < 0.01$), or about a 13% shortfall; the PPML count model gives −0.419, equivalent to a $\sim 34\%$ multiplicative shortfall—closer to the raw aggregate −25% soccer drop and −17% volleyball drop. The two estimators bracket the magnitude: OLS underestimates because $\log(1 + x)$ compresses small counts; PPML overestimates if there is genuine residual heterogeneity across country-sport cells.

After 2020 both estimators show non-positive but small-and-insignificant coefficients in 2021–2023, consistent with recovery in international soccer and volleyball recruitment. The static DiD (averaging post-2020 effects) is −0.033 in OLS (*n.s.*) and −0.128 in PPML ($p < 0.01$), reflecting that the PPML picks up a small persistent drag after 2020 in addition to the 2020 spike. Figure ?? plots the OLS leads/lags.

Heckman correction matters. The selection probit (extensive margin: $\Pr(\text{any athletes in country} \times \text{year} \times \text{sport})$) yields a large positive inverse Mills ratio in the intensive equation, +1.81 ($p < 0.05$). After correcting for selection, log GDPpc enters at +0.68 ($p < 0.10$), political stability returns to +0.32 ($p < 0.10$), and the population elasticity stays at +0.28 ($p < 0.05$). The Heckman intensive estimates are larger than the pooled OLS coefficients but have wider standard errors. We read this as: there is real selection into the country-sport-year cells that are positive, and after correcting for it the GDP and stability margins are detectable but weaker than the cross-sectional story.

3.7 Region fixed effects horse-race

To quantify how much of the unexplained variation in country-panel supply is regional-pipeline-shaped, we add eight-region fixed effects to the count specification.

	Macro only	Macro + 8 region FE
log pop _{15–24}	+0.35*** (0.07)	+0.40*** (0.07)
log GDPpc	+0.37** (0.16)	+0.53*** (0.18)
Political stability	+0.38 (0.24)	+0.38 (0.24)
R^2	0.21	0.32

Adding region FE raises R^2 from 0.21 to 0.32, a meaningful but not dominant gain. A specification with region FE only (no macro covariates) achieves $R^2 = 0.13$. Read together: macro covariates

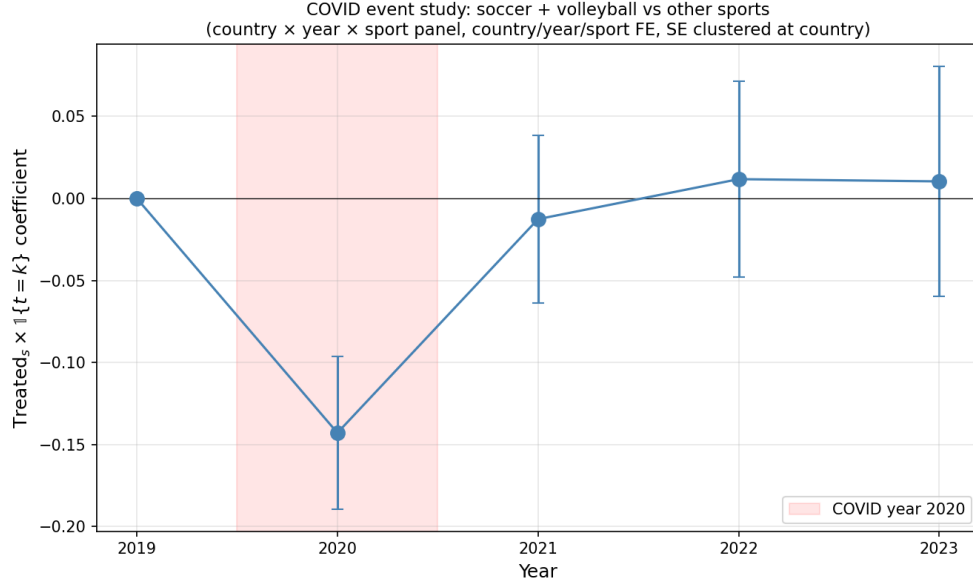


Figure 1: COVID event study on the country $\times$ year $\times$ sport panel. Each coefficient is $\text{Treated}_s \times \mathbb{1}\{t = k\}$ where $\text{Treated}_s = 1$ for soccer and volleyball, 0 otherwise. Base year 2019. $n = 6,040$ cells, 152 countries, 5 years, 8 sports. Country, year, and sport fixed effects; SEs clustered at country.

and regional pipelines each contribute, in roughly equal share, to a total explained variance of about a third. The other two-thirds remain unexplained even with both—consistent with the discussion in Section ?? that country-specific (not region-specific) pipeline structure dominates.

3.8 Home-country shocks and athlete retention

The country panel measures *entry*: how many international athletes arrive in a given country-year. A complementary margin is *persistence*—once an international athlete is on a U.S. NCAA roster, do shocks at home affect whether they remain enrolled? And do those shocks spill over to their domestic teammates? We use the 2019–2023 athlete-year roster panel (8 sports, $N=260,737$ athlete-years) and define the outcome as whether the athlete appears on the same school’s roster in year $t+1$.¹ Mean retention is 75.1%; the gap between international (74.7%) and domestic (75.2%) is negligible unconditionally.

Own-shock effects. We regress $\text{Pr}(\text{return}_{i,t+1})$ on a shock indicator for athlete i ’s home country in year t , including year and sport fixed effects, with standard errors clustered at the country level.

¹Athlete identifier is the school $\times$ sport $\times$ gender $\times$ name string. Name collisions are possible but rare within a school $\times$ sport. The sample is restricted to under-classmen so that scheduled graduation does not dominate the variation.

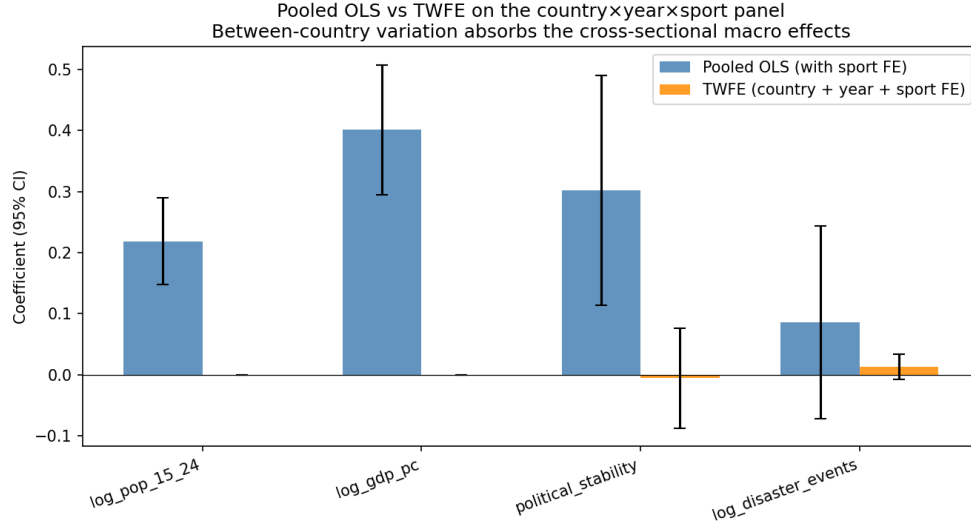


Figure 2: Pooled OLS vs. TWFE coefficient comparison on the long panel. Bars are point estimates with 95% confidence intervals (country-clustered SEs). The political-stability and log-GDPpc coefficients collapse to zero once country fixed effects are included, indicating they capture between-country composition rather than within-country variation.

Home-country shock	β on $\Pr(\text{return})$		N
GDP shock ($z < -1.5$)	-0.002	(0.026)	19,921
Currency crisis	-0.004	(0.024)	19,921
Severe disaster	-0.006	(0.011)	19,921
Armed conflict (UCDP)	-0.033**	(0.014)	19,921
Political-stability drop	-0.034**	(0.015)	19,921

Pure macroeconomic shocks (GDP, currency, unemployment) do not move international retention. *Security* shocks do: a conflict episode or a sharp political-stability decline at home reduces same-school retention by 3.3–3.4 percentage points, about a 4.5% relative drop from the 75% baseline. The two security indicators are partially correlated but not redundant; running them jointly yields very similar magnitudes.

Sport heterogeneity. The conflict and political-stability effects are concentrated in the sports with the largest international intakes and the broadest country mix:

	Conflict β	Polstab-drop β
Soccer ($N=6,125$)	-0.116***	-0.092***
Track ($N=3,187$)	-0.056***	-0.054***
Volleyball ($N=932$)	-0.063**	-0.073**
Ice Hockey ($N=1,315$)	-0.098**	—
Basketball, Swim, Tennis	n.s. in either column	

Soccer is the largest in absolute magnitude: an athlete whose home country experiences armed conflict in year t is 11.6 pp less likely to remain on the same U.S. roster the next year—roughly one in nine soccer players exits at the margin.

Peer spillovers. If a shock disrupts an international athlete enough to change their retention, does it also affect their domestic teammates? We construct, for each domestic athlete-year, the exposure

$$\text{Expo}_{i,t} = s_{(s,t)}^{\text{intl}} \cdot \bar{D}_{(s,t)}^{\text{shock}}$$

where $s_{(s,t)}^{\text{intl}}$ is the share of athlete i ’s team that is international and $\bar{D}_{(s,t)}^{\text{shock}}$ is the average shock indicator across those international teammates’ home countries. The coefficient is identified from cross-school variation in roster composition interacted with which countries were shocked when.

Exposure of domestic athlete to ...		β	N
Currency-crisis teammates	+0.052	(0.120)	122,513
GDP-shocked teammates	+0.020	(0.045)	122,513
Conflict-affected teammates	− 0.209 **	(0.100)	122,513
Politically-unstable-home teammates	− 0.267 ***	(0.104)	122,513

The same two shocks that move international athletes’ own retention also move their domestic teammates’. The coefficients are scaled to “fully exposed” teams ($s^{\text{intl}}=1$, all from shocked countries); a more realistic 20% international roster with half from a politically-unstable country ($\text{Expo}=0.10$) implies a 2.7 pp drop in domestic retention—about half the direct effect on the shocked athletes themselves.

Mechanism interpretation. Three patterns are consistent with the data: (i) the responsive shocks are those that disrupt *contact with home* (conflict, regime instability) rather than disrupting *financial flows* (GDP, currency); (ii) the spillover to teammates is consistent with team-level disruption (training time lost, coaching attention redirected, roster instability mid-season); (iii) the effect concentrates in the team sports with the largest international rosters. We cannot separately identify discrete mechanisms—visa complications, family emergencies, mental health, transfer to another U.S. school—because the data record only roster presence at one school, not destination of departure. A fraction of the “exits” may be intra-NCAA transfers rather than withdrawals; we treat this as a measurement caveat rather than a direction-of-bias concern.

What we do not estimate. The dependent variable is roster persistence, not academic performance. Direct grade or graduation outcomes (APR, GSR) would sharpen the interpretation but are publicly available only at the team-sport-school-year level and were not collected here.

4 Discussion

Four substantive conclusions emerge.

First, **wealth and population scale govern entry into the international NCAA pipeline.** Both $\log \text{GDPpc}$ and $\log \text{pop}$ enter the extensive-margin probit at $p < 0.01$; political stability does not. The mechanism is the fixed-cost story familiar from migration economics: producing a recruitable athlete requires some minimum infrastructure and discovery-cost expenditure that scales with national resource flow.

Second, **conditional on entry, athletes scale sublinearly with cohort population.** The estimated elasticity is between 0.33 (OLS) and 0.61 (Heckman two-step), with the selection-corrected

estimate being our preferred reading. Doubling a source-country’s 15–24 cohort raises NCAA supply by roughly $1.5\times$, not $2\times$. The most plausible mechanism is finite US-side recruitment capacity: the marginal NCAA program does not scale its international intake one-for-one with any single source country. Small wealthy countries are therefore over-represented per capita relative to populous ones. Quantile regression refines this: the elasticity is 0.24 at the low end of supply and 0.51 at the high end, so the diminishing-returns story is strongest for small-supply countries.

Third, **political stability has an intensive-margin effect of uncertain magnitude**. OLS gives a statistically zero estimate of +0.37 (SE 0.24), but with $n = 100$ the minimum detectable effect at 80% power is 0.68, so the data do not strongly reject moderately positive effects. The Negative Binomial estimate is +0.79 ($p < 0.01$); the Heckman estimate is +0.60 ($p < 0.10$). We cannot pin down whether stability matters at the country-pool level, because the country pool averages over very heterogeneous sport structures. *Within* distance running, however, political stability is large and significant (+0.93, $p < 0.05$): the East African pipeline depends on it. Within sprints, it does not. Within throws and jumps, sample sizes preclude a strong claim.

Fourth, **the relevant unit of analysis may not be the country**. Adding eight region fixed effects raises R^2 from 0.21 to 0.32 in the count specification; the macro covariates’ point estimates barely move. Region absorbs some but not most of the unexplained variance. The sport-event split is more informative than the regional split. Both point to the same conclusion: the country-panel pooled regression is averaging over qualitatively different supply curves (Caribbean sprint network, East African distance network, European throwing pipelines, etc.) that share little structure beyond GDP and population. A complete model would estimate gravity equations at the sport-event level with bilateral US-source-country variables (recruiting links, scholarship agreements, language commonality, alumni networks) that we do not have in our data.

5 Caveats and limitations

- **Cross-section, not panel.** Roster snapshots are 2024–25 / 2025–26; macro covariates are 2023 and 2010–23 aggregates. A descriptive pseudo-panel based on athletes’ class-year gives at most five entry cohorts with no clean pre-period for any shock event.
- **Coverage.** Hometown is observed for 87.6% of track athletes; the unobserved 12.4% are missing not at random (JS-rendered roster pages that systematically affect a few major D-I programs).
- **Track-and-field dominance.** 96% of the international sample competes in track. The paper’s findings should be read as describing the international track-and-field market specifically.
- **Cohort denominator approximation.** We use the average of the 15–24-year-old cohort 2018–2022 as the denominator. The ideal denominator would be 18–22 to match enrolled ages; we attempted this but the World Bank does not provide aggregate 18–22 totals. Rescaling 15–24 by a constant fraction is observationally equivalent in the count model (the constant is absorbed by the intercept), so this does not change our slope estimates.
- **Heckman identification by functional form.** The exclusion restriction in our Heckman two-step is the probit nonlinearity, not a covariate-level exclusion. The IMR coefficient (+3.27, SE 2.64) is large but imprecisely estimated, so selection-correction matters but its magnitude

is uncertain. A proper exclusion (e.g. a US-side measure of recruitment intensity) would improve identification.

- **Modest R^2 .** Macro + region explain about a third of intensive-margin variance; the residual is most plausibly driven by country- and sport-specific recruitment pipelines that we cannot proxy.

6 Reproducibility

All results in this paper are reproducible from local Python scripts:

- Roster scraping: `discover_track_urls.py`, `run_track_scrape.py`, `run_track_selenium.py`, `rediscover_urls.py`.
- Country mapping: `run_country_mapping.py`.
- Macro merge: `run_merge_analysis.py`, `fetch_extra_shocks.py`, `run_join_shocks.py`.
- Main regressions: `run_main_results.py` (extensive probit + intensive OLS chain).
- `run_revisions_2.py`: Heckman two-step, MDE for political stability, quantile regression at $\tau \in \{0.25, 0.50, 0.75\}$, Negative Binomial with free log pop, and region FE horse-race.
- `run_event_analysis.py`: expanded sport-event analysis (composition, regional shares, within-event OLS and NegBin for all eight event groups, country-event specialisation residuals) — Section 3.5.
- `run_historical_scrape.py`, `run_historical_scrape_multisport.py`: 2019–2023 roster panels for eight sports (track, soccer, basketball, tennis, golf, swimming, ice hockey, volleyball). Uses the student-built `school_website_platforms_full.csv`.
- `build_long_panel.py`: stacks the eight sport CSVs into the country $\times$ year $\times$ sport long panel and merges time-varying WGI, WDI, UCDP, and EM-DAT covariates.
- `run_panel_specs.py`: produces the four-spec panel results (pooled OLS, TWFE, COVID DiD with leads/lags, Heckman two-step on the panel) reported in Section 3.6.
- `make_panel_plots.py`: produces `output/fig_event_study_panel.png` and `output/fig_pooled_vs_twfe.png`.
- `run_stress_tests.py`: drop-top-5 disaster countries, events-vs-deaths comparison, refugees raw count.
- `run_bartik.py`, `run_event_study.py`, `run_paper_outputs.py`: auxiliary results referenced in Sections 3.6 and 5.

Figures

References

Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates.

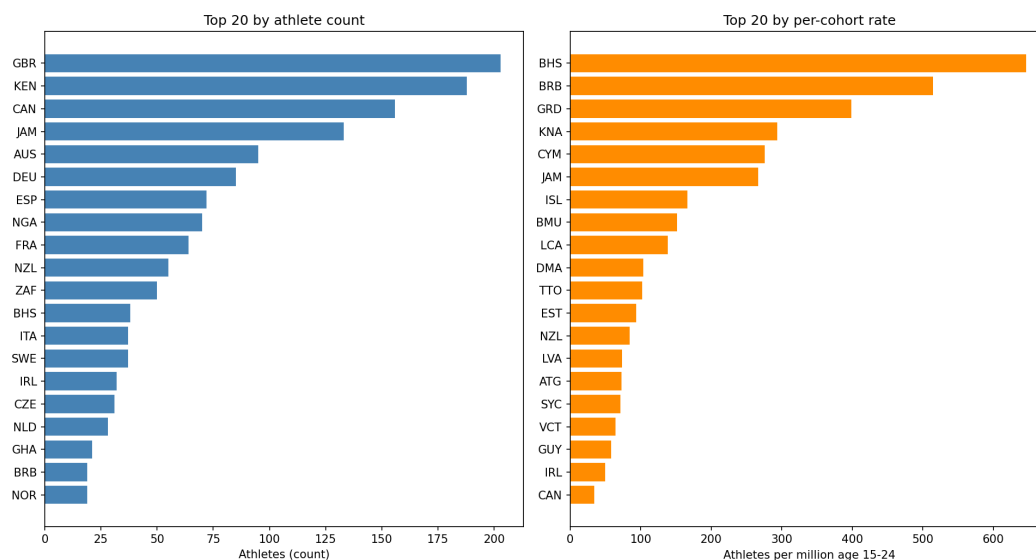


Figure 3: Top 20 source countries by raw athlete count (left) and by athletes per million 15–24-year-olds (right).

Lee, D. S., J. McCrary, M. Moreira, and J. Porter (2022). “Valid t -Ratio Inference for IV.” *American Economic Review* 112(10): 3260–3290.

Santos Silva, J. M. C., and Silvana Tenreyro (2006). “The Log of Gravity.” *Review of Economics and Statistics* 88(4): 641–658.

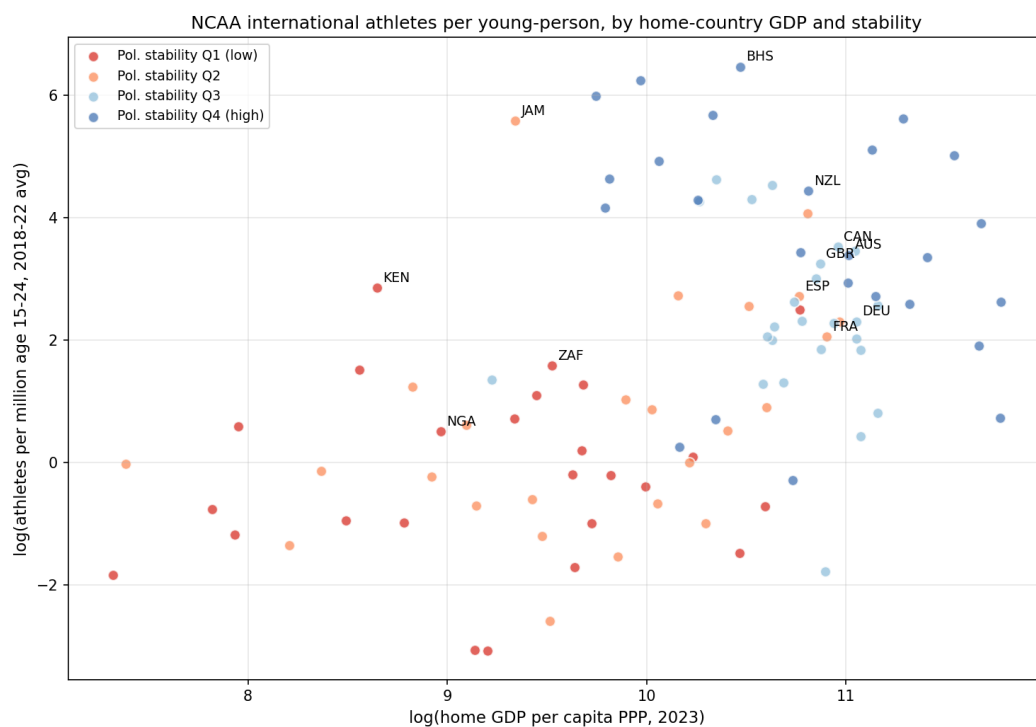


Figure 4: Log athletes per million age 15–24 vs. log home-country GDP per capita, colour-coded by political-stability quartile. The apparent stability gradient is largely a country-size artefact in this view; see Section 3.3.